

Al Hotimski

Geospatial Data Scientist | a.hotimski@gmail.com | LinkedIn: [in/al-hotimski](#) | GitHub: [ahotimski00](#)
| Portfolio: [ahotimski00.github.io](#)

Education

Boston University, MS Remote Sensing and Geospatial Sciences (Environmental Data Science), Dean's List, BA Earth and Environmental Science (Computer Science Minor) (2021-2025)

Coursework: Advanced GIS, Remote Sensing, Data Structures, Python Programming, Statistical Modeling, Environmental Science

Experience

Geospatial Data Scientist, The Conservation Fund (*Internship Jun-Dec 2025, Full-time Mar 2026-Present*)

- Built an automated USDA NRCS ACEP easement eligibility screening tool (Python / ArcGIS Pro) capable of regional-scale assessment that NRCS's own tools cannot perform; **reduced analysis time from 20+ hours to 10 minutes (>99%), projecting 240 hours and \$12,000/year in savings** at monthly cadence
- Designed a dynamic geodatabase tracking land parcelization, boundary changes, and ownership transfers across U.S. timberland; built a custom fuzzy-matching package (performance-critical components in C-based Python) to resolve owner name inconsistencies at scale; **reduced manual QA from ~2 weeks to under a day (93%)**
- Built a child-to-parent LLC hierarchy database enabling outreach at the parent company level rather than individual subsidiaries; built a multi-timestamp ingestion pipeline flagging boundary and ownership changes to track consolidation, divestment, and fragmentation trends
- Building Salesforce (Locate) to ArcGIS Online integration to automate real-time spatial layer sync, replacing a fully manual project approval and intake workflow and giving conservation staff direct access to live project boundaries
- Applied CCDC (Continuous Change Detection and Classification) across ~20,000 acres in Google Earth Engine to map historical forest disturbance for carbon accounting; produced enterprise-scale spatial datasets used to prioritize timberland acquisition and stewardship decisions

Research Assistant, PLACES Lab, Boston University (*May-Aug 2025*)

- Validated and interpreted 800+ georeferenced locations spanning 25 years of land cover change; developed QA/QC protocols and review checklists that improved classification consistency across the team
- Resolved classification conflicts and data anomalies in collaboration with senior researchers

Research Project Team Manager, Boston University / Boston Housing Authority (*Sep-Dec 2024*)

- Led GIS analysis for a public-sector client assessing Urban Heat Island exposure across Boston Housing Authority properties

- Integrated spatial datasets (land cover, surface temperature, exposure metrics) to identify high-risk locations; developed ArcGIS Online web maps and visual summaries for non-technical stakeholders
- Gathered requirements from BHA staff, scoped deliverables, and presented findings directly to BHA staff, translating spatial analysis into planning recommendations they could act on

Environmental Data Science Internship, TRC Companies, Inc. (*May-Aug 2024*)

- Developed Python automation scripts to streamline data entry and QA/QC processes across internal GIS and reporting workflows
- Performed spatial and tabular data validation across multi-source GIS inputs used in client deliverables, ensuring accuracy and consistency per TRC internal standards

Research Team Lead, Boston University UROP (*Feb-Aug 2022*)

- Secured two competitive undergraduate research grants; led a four-person team through a full feasibility analysis of electricity cost optimization in controlled farming environments, including cost modeling and regulatory review
- Presented findings at a university research symposium

Skills

Languages: Python · R · SQL · JavaScript · Bash

GIS & Remote Sensing: ArcGIS Pro · ArcGIS Online · QGIS · Google Earth Engine (cloud-native) · geopandas · shapely · rasterio · Landsat

Machine Learning & Analysis: scikit-learn · supervised classification · fuzzy matching · data harmonization · entity resolution · time-series · change detection (CCDC)

Data Engineering: pandas · NumPy · ETL pipelines · relational databases · REST APIs · Git · Linux

Visualization & Reporting: Power BI · ArcGIS Online dashboards · matplotlib

Volunteer

Girls Who Code, Instructor (*2020-2021*) Taught introductory programming to students from underrepresented communities.

Family Service of the Lehigh Valley, Volunteer (*Ongoing*)

Certifications & Coursework

Certifications: OSHA 40-Hour HAZWOPER · Massachusetts Boating License

Coursework: Remote Sensing and Digital Image Processing · Advanced GIS and Spatial Analysis · Statistical Modeling for Environmental Data · Machine Learning (Python / scikit-learn)